

Processing and memory

Microchips process and control the flow of data and instructions inside a computer. They interact with other parts of the computer to produce outputs.

SEE ALSO

◀ 34–35 Inside a computer

◀ 36–37 Peripheral devices

What is hardware?

48–49 ▶

Central Processing Unit (CPU)

The CPU is where all the work of the computer is done. It is made up of a control unit (CU), registers (which are temporary places in the CPU where values can be stored), and an arithmetic logic unit (ALU). The CPU has a fetch-decode-execute cycle: one instruction is fetched from memory by the control unit and translated into binary numbers, which are stored in registers. These numbers are passed into the ALU, which executes the logical or arithmetic operations necessary. Modern computers often have more than one processing unit, called a core, in the CPU.

IN DEPTH

Von Neumann architecture

The organization of computer components is known as a “von Neumann architecture”. It’s named after John von Neumann (1903–1957), a Hungarian-American physicist and mathematician, who described it in a report on the EDVAC (Electronic Discrete Variable Automatic Computer) computer in 1945.

